

Week1 Session 1

1. Linux Shell(command line)

1.1 Directory Command (目录命令)

command	Action	desc
mkdir [dir_name]	增	makes a new (empty) named directory, (eg. mkdir myCode)
rmdir [dir_name]	删	remove an empty named directory, (eg. rmdir myCode),只有空目录才可以删除，目录中有文件和子目录则失败
pwd	查	tells you your current directory (folder)
ls	查	list everything in the current directory (files or directories)
cd [dir_name]	改	move into a named directory (eg. cd myCode)
cd ..	改	move up one directory level

1.2 File Command (文件命令)

command	Action	desc
touch [file_name]	增	makes a new (empty) file (eg. touch myFile.txt)
cp [src_file] [dest_file]	增	copy src_file to dest_file
rm [file_name]	删	remove a file (eg. rm myFile.txt)
cat [file_name]	查	display the content of the text file
rename [file1] [file2]	改	reame file1 to file2

1.3 扩展阅读

[Linux命令大全](#)

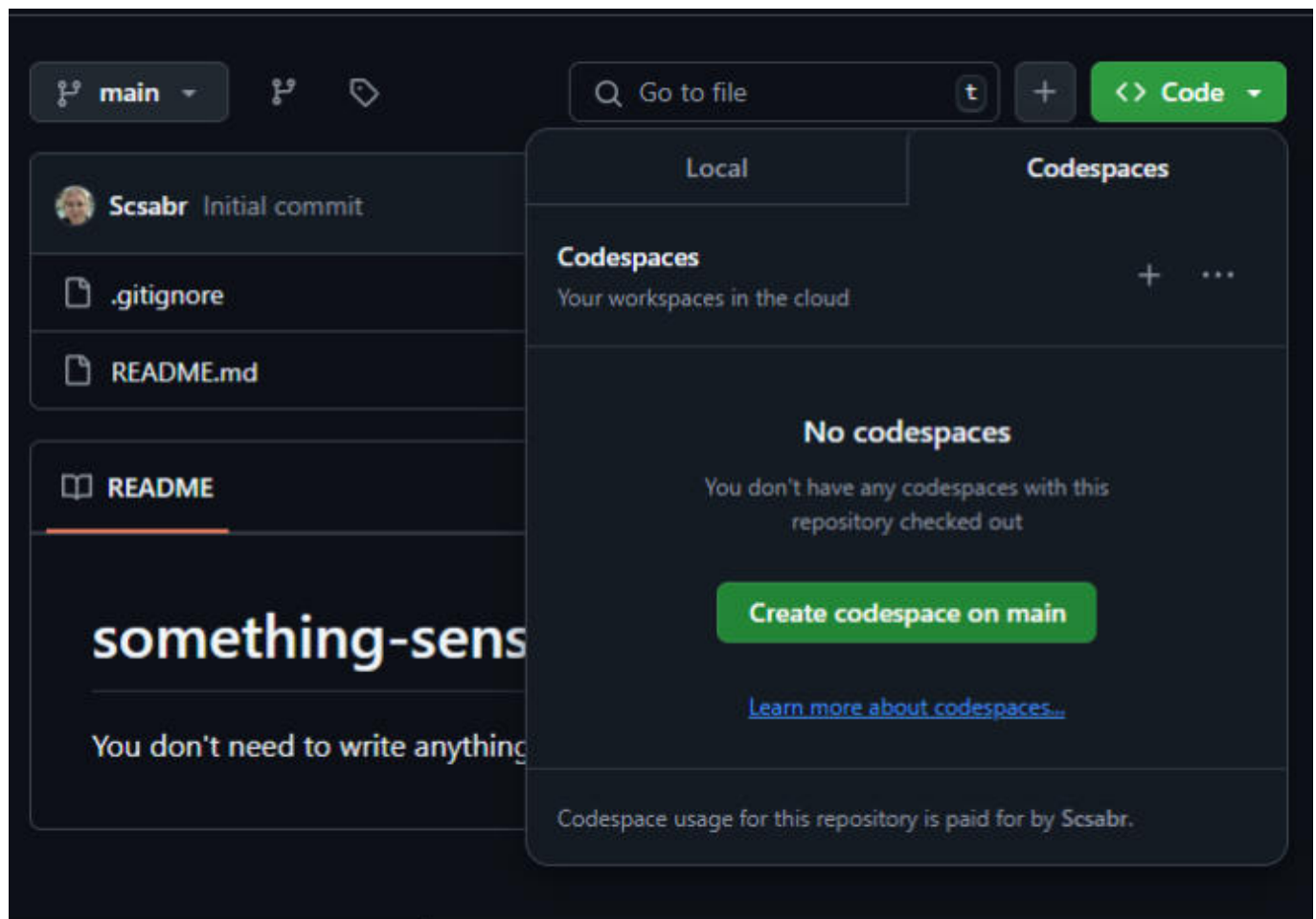
2. GitHub

- A git-based platform for version control
- Also has lots of cool open source projects!
- We're going to be using it for everything

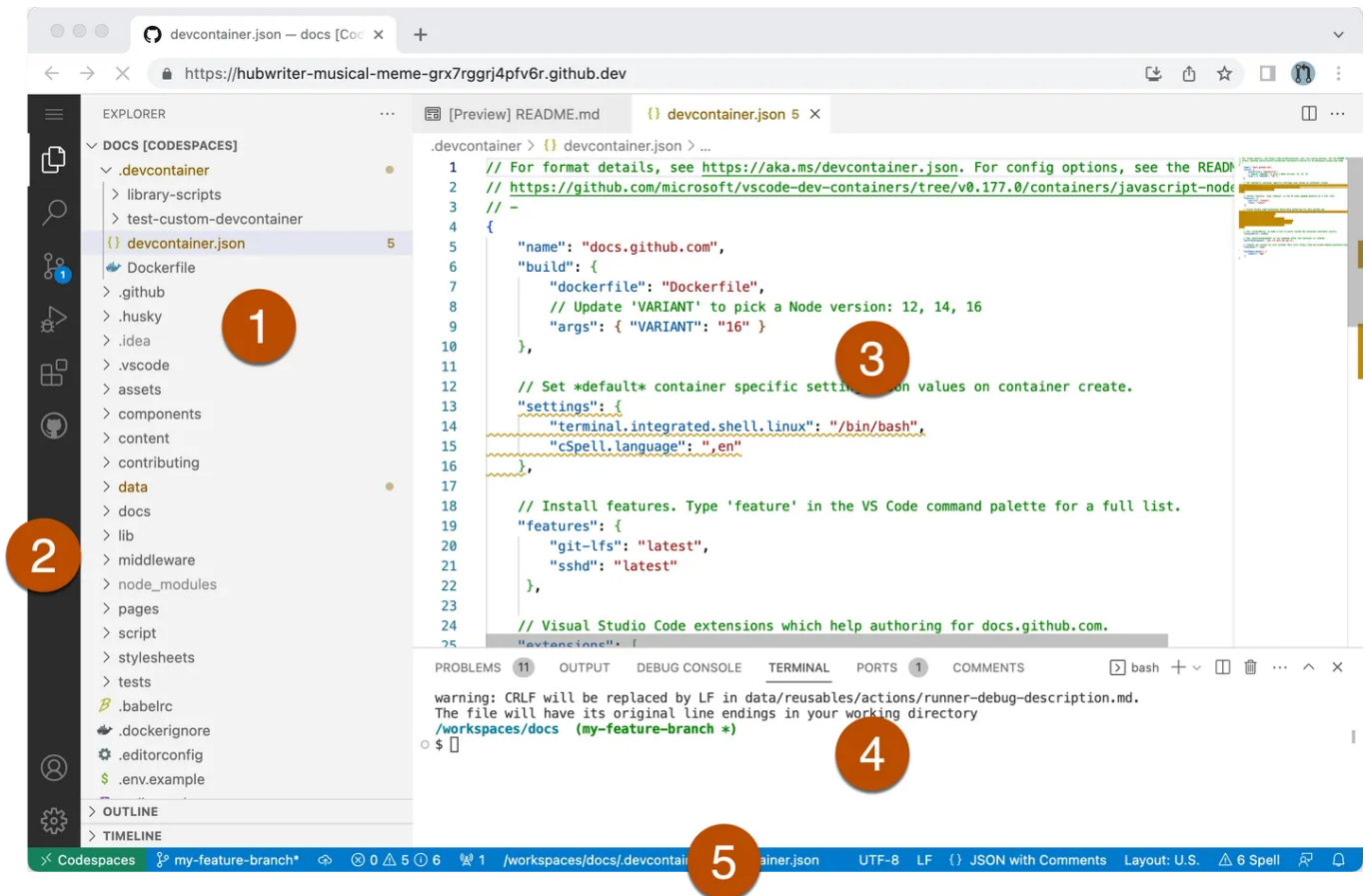
3. Codespaces

- integrated development environment inside Github (IDE:集成开发环境)

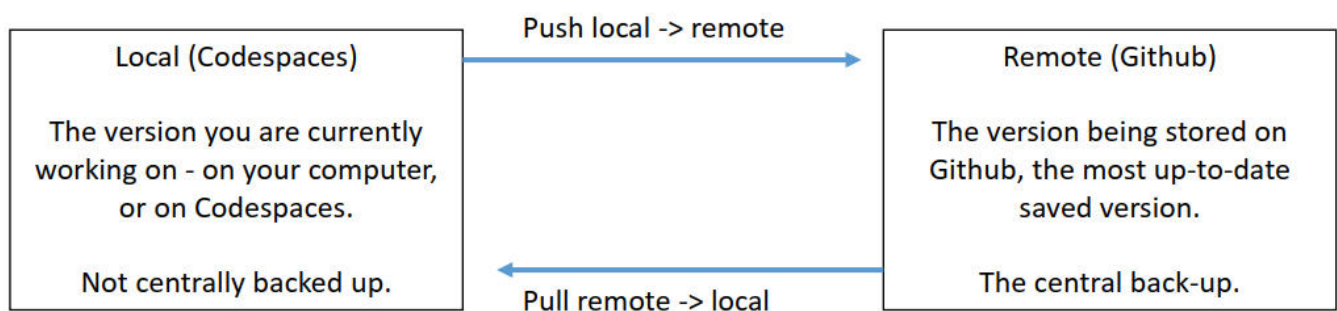
3.1 Open a Codespaces



3.2 What is a Codespace?



- A Codespace is a ****linux virtual machine**** running in the cloud. (运行在云上的linux虚拟机)
- It is created with the current 'main' branch from the Github repository. (基于main分支创建)
- You can run, edit and write new code inside the Codespace. (在Codespace中编写运行代码)
- However your work is only saved locally inside the Codespace. (代码保存在Codespace虚拟机中)
- You can push your work to save it permanently (想永久化保存代码，需push到github仓库中)



3.3 Why Codespace

- Some of you are programming on Windows machines, some on Mac, and some on Linux.
- Codespaces means you all work on the same machines. (意味着你们都在同一台机器上工作)
which is helpful to ensure your work is portable
- Being able to work on Linux is also essential. (能够在Linux上工作也很重要)
if you work as a developer, it is almost guaranteed that you will use either linux or another Unix-based system (such as a Mac).

3.4 扩展阅读

[GitHub Codespaces 快速入门](#)

4. Git

Git is a version control tool(版本控制工具)

4.1 Why do we need a version control tool ?

In industry

- You are always working in a team
- You work on the same codebase (代码库) as other people
- The codebase may be live(already deployed and running)

In university

- Working individually or in teams
- You may be working on university machines or your own
- Often creating new code
- You need a secure backup of your work
- You will break things and want to recover working code!

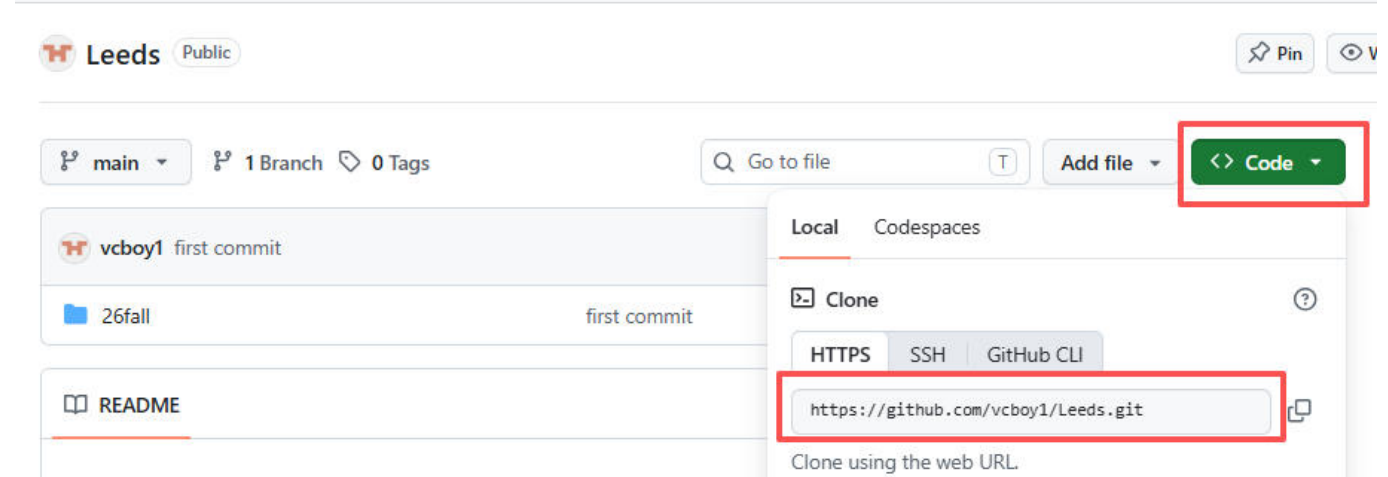
4.2 The role(作用) of version control system(git)

- allow programmers to work concurrently (并行工作) on the same code.
- allow you to track the history of development. (跟踪开发历史)
- allow you to restore previous versions (恢复以前的版本)

4.3 将github仓库clone到本地工作空间(只做一次)

- 如用CodeSpace打开项目，则无需此步骤
- 仓库只用clone一次，后续用pull同步

步骤	说明
cd /src	进入根目录下的src目录
git clone https://github.com/vcboy1/Leeds.git	将仓库clone到本地的/src目录下，代表仓库的.git路径获取方式见下图
cd Leeds	clone成功后，进入本地底仓目录Leeds
git log	查看修改历史



4.4 配置提交记录里的作者身份

- 如用CodeSpace打开项目，则无需此步骤
- 这是全局设置，只做一次，设置后无需再设置

步骤	说明
git config --global user.name "Xinyuan Li"	设置提交者名字
git config --global user.name "利兹邮箱"	设置提交者邮箱

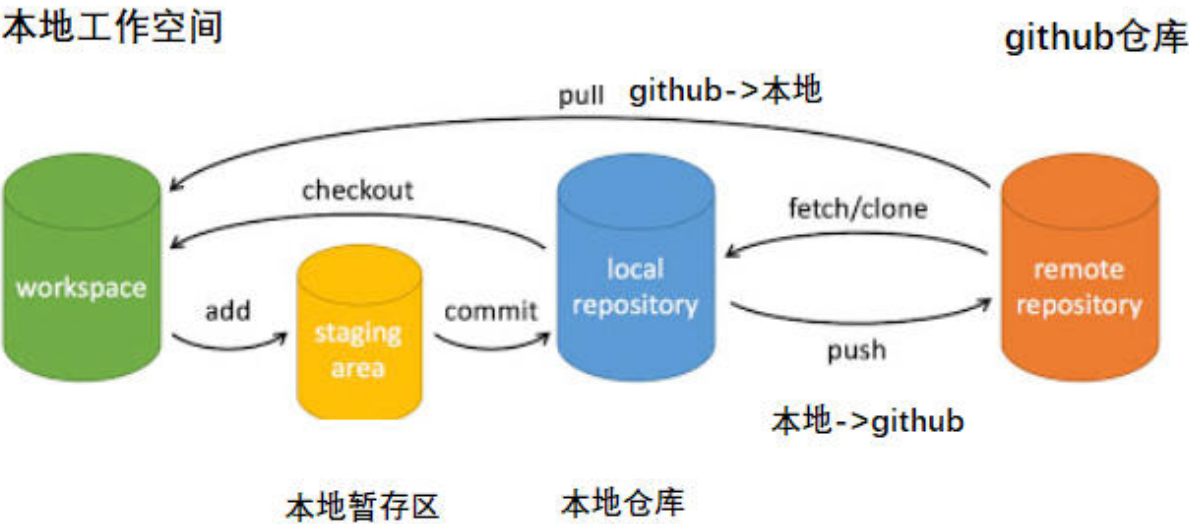
4.5 日常操作：从github仓库拉取最新代码到本地工作空间

- 在每次开发前，首先要pull最新代码，将本地工作空间的代码和远程仓库同步，保证开发的是最新的代码。

步骤	说明
git pull origin	从远程origin仓库拉取最新代码到本地工作空间

4.6 日常操作：从本地工作空间的修改推送到github仓库

工作原理



- 1. pull,clone将github远程仓库的代码拉到本地仓库
- 2. 在本地workspace中修改 的文件，通过git add 命令加入到本地暂存区
- 3. 通过git commit命令将本地暂存区的多个修改文件，一次打包成一个commit，并给一个备注-m。备注的意义是为了让其他开发者知道，这个commit是干什么的，比如：修复bug1021, 或者新增一个功能201.
- 4. 打包后的commit存在本地仓库，需要用git push 把commit推送到github仓库
- 5. 其他开发者pull时，他们的本地仓库就会包含你提交的commit代码了，这就是团队开发

步骤	说明
git pull origin	从远程origin仓库拉取最新代码到本地工作空间
	做正常的开发工作
git add .	stages(暂存) your changes – it tells Git that you have done some work you would like to save
git commit -m "xxx"	bundles(打包) these changes into one 'commit' (提交) you can go back to previous commits, so this is basically a way of creating a save point.(你可以回滚到以前的提交，这是一个创建还原点的方法，一个commit就是一个还原点
git push origin	sends these changes back to the remote Github server(把commit打包的changes push到github 仓库中)

4.7 扩展阅读

- [Git命令行基本操作](#)
- [Git GUI基本操作](#)

5. MarkDown Format

- Documentation is important on Github
- Markdown is a lightweight markup language (like HTML) which is used to write documentation and other basic text documents.
- It uses simple ‘tags’ such as # and - to format text.
- markdown or “md” files are often used for READMEs.

5.1 章节(#)

```
# 号和标题文字之间必须有一个空格
# 一级标题
## 二级标题
### 三级标题
#### 四级标题
##### 五级标题
##### 六级标题
```

一级标题

二级标题

三级标题

四级标题

五级标题

六级标题

5.2 普通文本

普通文本直接输就可以了。
如果要换行使用两个以上空格加上回车。

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5.3 列表

无序列表使用加号(+)作为列表标记，这些标记后面要添加一个空格，然后再填写内容

- + 第一点
- + 第二点
- + 第三点

- 第一点
- 第二点
- 第三点

5.4 忽略tag的文本块

移到空行首部，按tab，然后输入文字，此时文字都是无格式了

5.5 超链接

文字超链接： [文字提示](超链接地址)
图片超链接： ![文字提示](超链接地址)

文字超链接



5.6 表格

	表头		表头	
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	单元格		单元格	
	单元格		单元格	

表头	表头
单元格	单元格
单元格	单元格

5.7 分割线

本文件的Markdown版本在:

https://github.com/vcboy1/Leeds/blob/main/26fall/1850/Note/w1s1_linux_github_markdown.md